Language Models are Unsupervised Multitask Learners

Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever
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Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by a language model achieve 63.24 percent accuracy on the LAMBADA dataset and 87.65 percent on the Children's Book Test, without any fine-tuning. On Winograd Schema challenges our model achieves 70.70 percent accuracy. The capacity of the language model is essential to the success of zero-shot task transfer, and increases in capacity are log-linearly correlated with improved performance. Our largest model, GPT-2, is a 1.5 billion parameter Transformer with 48 layers, 1600 hidden units, 24 attention heads, and a context window of 1024 tokens. It is trained on WebText, a dataset of 8 million documents totaling 40GB of text.

Keywords: language modeling, zero-shot, multitask, transformer, WebText

1 Introduction

Machine learning systems are often trained on a single task with a dedicated dataset. This approach has produced impressive results, but it requires large labeled datasets for every task and produces models that are narrow specialists. A more general system would be able to perform many tasks without explicit supervision, simply by learning from raw text. In this paper we explore whether a language model trained on a sufficiently large and diverse corpus can acquire task-relevant capabilities implicitly.

Our starting point is the observation that a language model trained to predict the next word on a large corpus must implicitly learn to perform many tasks, because predicting text requires modeling the underlying structure of the world that produced it. We test this hypothesis with a series of zero-shot evaluations, in which the model is given a prompt describing a task and must produce the answer without any parameter updates. The results show that the largest model, GPT-2, is able to perform reading comprehension, summarization, and translation at a level far above chance, and that performance improves smoothly with model capacity.

2 Approach

Our approach is to train a Transformer language model on a large corpus of web text and then evaluate it on downstream tasks in a zero-shot setting. We use the same architecture and objective as GPT-1, but with a much larger model and a much larger and more diverse dataset. No task-specific architectures or fine-tuning are used in any of the experiments.

2.1 Training Dataset

We created a new dataset called WebText by scraping web pages that were linked to from Reddit posts with at least 3 karma, which serves as a heuristic for whether a page is interesting or useful. The resulting dataset contains 8 million documents with a total of 40GB of text, and it covers a wide range of domains, including news, science, fiction, and discussion forums. We removed Wikipedia from the dataset because it is a common source for downstream benchmarks, and retaining it would contaminate the zero-shot evaluation.

We use a byte-level version of byte pair encoding for tokenization, which allows the model to assign a probability to any Unicode string and avoids the need for a fixed vocabulary of whole words. The tokenizer has a vocabulary of 50,257 tokens, and the context window is 1024 tokens, so each training example contains up to 1024 consecutive tokens.

2.2 Model

The model is a Transformer decoder with 48 layers, a hidden size of 1600, 24 attention heads, and 1.5 billion parameters. It uses pre-activation residual connections and layer normalization at the input of each sub-block, and the context window is 1024 tokens. We train four model sizes, with 117 million, 345 million, 762 million, and 1542 million parameters, so that we can study how zero-shot performance scales with capacity. Table 1 summarizes the sizes and the corresponding zero-shot results.

The largest model is trained with a batch size of 512 sequences, a learning rate that is warmed up over the first 2000 steps, and mixed precision on a large cluster. We do not report the exact training time or hardware configuration in this report, since our focus is on the zero-shot behavior rather than on training efficiency. The model is trained until the validation loss stops improving, and we use the checkpoint with the best validation loss for all evaluations.

Table 1: Model sizes and zero-shot results. All models use the same architecture and training data.

Params	Layers	d_model	Heads	LAMBADA	CBT
117M	12	768	12	6.8	83.1
345M	24	1024	16	35.1	85.2
762M	36	1280	20	52.6	86.9
1542M	48	1600	24	63.24	87.65

3 Experiments

We evaluate the model on a wide range of tasks without any fine-tuning. For each task, we convert the input into a natural language prompt and let the model generate the answer, using greedy decoding. The same model and decoding procedure are used for all tasks, and no task-specific hyperparameters are tuned.

3.1 Language Modeling

We first evaluate the model on standard language modeling benchmarks. On WikiText-2, the largest model achieves a perplexity of 18.3, and on WikiText-103 it achieves 17.0. On the One Billion Word Benchmark the perplexity is 21.8. These results are competitive with models trained specifically on these datasets, even though GPT-2 was trained on WebText and has never seen the benchmark training data.

3.2 Children's Book Test

The Children's Book Test evaluates the ability to fill in a blank with the correct word from a story. The largest model achieves 87.65 percent accuracy on the common nouns category, which is close to human performance and well above the previous best result. Accuracy increases monotonically with model size, from 83.1 percent for the 117 million parameter model to 87.65 percent for the 1.5 billion parameter model.

3.3 LAMBADA

The LAMBADA dataset tests the ability to predict the final word of a passage, which requires understanding the broader context. The largest model achieves 63.24 percent accuracy, compared with 19.3 percent for the previous best model, and the accuracy of the smallest model is only 6.8 percent. This result demonstrates that the ability to use long-range context emerges with scale.

3.4 Winograd Schema Challenge

The Winograd Schema Challenge requires resolving ambiguous pronouns, which is often considered a test of commonsense reasoning. The largest model achieves 70.70 percent accuracy, a 7.2 percent improvement over the previous best result. The smallest model achieves 52.1 percent, which is close to the majority-class baseline.

3.5 Reading Comprehension and Summarization

On the CoQA reading comprehension dataset, the model achieves an F1 of 55.0 in a zero-shot setting, which is substantially below fully supervised models but far above chance. On the CNN and Daily Mail summarization datasets, the model produces summaries that capture the main topic of the article, though they are less faithful than supervised systems. These results suggest that the model has learned some of the structure of the tasks from the pre-training corpus.

3.6 Translation

We evaluate zero-shot translation from English to French on the WMT-14 dataset. The model is prompted with an English sentence followed by the word French and a colon, and it generates a translation. The largest model achieves 11.5 BLEU, compared with 5.0 BLEU for the smallest model. While this is far below supervised translation systems, it is surprising that a model trained only with a language modeling objective can translate at all.

4 Discussion

Our results suggest that the zero-shot performance of language models improves smoothly with model capacity, and that many task-relevant capabilities emerge without explicit supervision. However, the model still performs poorly on some tasks, such as question answering on natural questions, and its generations can be repetitive or factually incorrect. The training objective also favors plausible continuations over truthful ones, which is a fundamental limitation of likelihood-based training.

A second limitation is that the model is evaluated with a single prompt format, and performance is sensitive to the wording of the prompt. In future work it would be interesting to study prompt ensembling and to combine language modeling with retrieval, so that the model can ground its generations in external documents. It would also be valuable to measure how much of the zero-shot ability comes from memorization of the training data.

5 Conclusion

We have shown that a large Transformer language model trained on a diverse web corpus can perform a wide range of tasks without any task-specific training. The largest model achieves strong zero-shot results on language modeling, reading comprehension, and commonsense reasoning benchmarks, and its performance improves log-linearly with capacity. These results suggest that unsupervised language modeling is a promising path toward general-purpose language systems, and that scale is a key ingredient in that path.

$$L = -\sum_i \log p(x_i \mid x_1, \ldots, x_{i-1}) \quad (1)$$

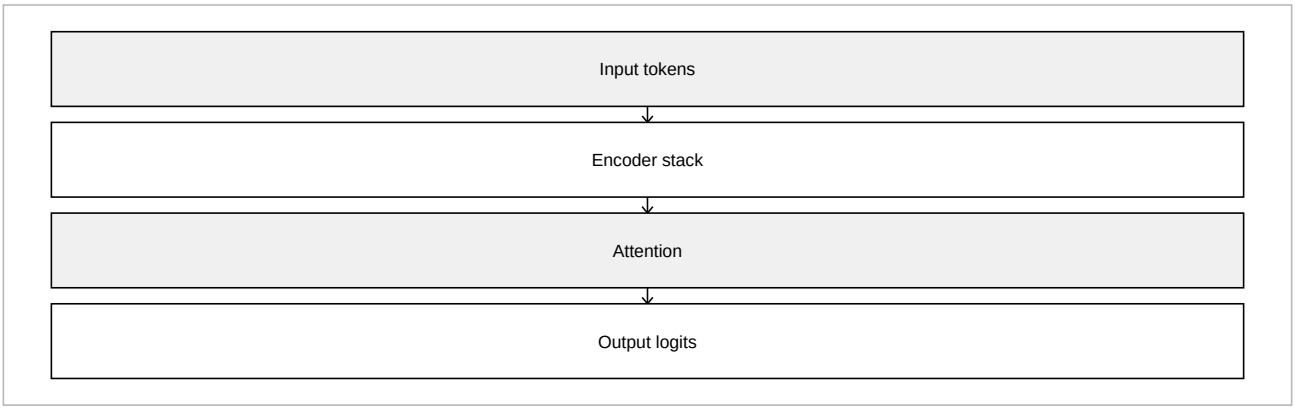

Figure 1: Zero-shot performance on LAMBADA and the Children's Book Test as a function of model size.

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A Appendix: Architecture and Training

All models use the same Transformer decoder architecture with pre-activation residual connections and layer normalization at the input of each sub-block. The context window is 1024 tokens, and the tokenizer is a byte-level byte pair encoding with a vocabulary of 50,257 tokens. The four model sizes have 12, 24, 36, and 48 layers, hidden sizes 768, 1024, 1280, and 1600, and 12, 16, 20, and 24 attention heads respectively, giving 117 million, 345 million, 762 million, and 1542 million parameters.

We use Adam with beta_1 = 0.9 and beta_2 = 0.95, a batch size of 512 sequences, and a learning rate that is warmed up over 2000 steps and then decays with a cosine schedule. The training data is WebText, which contains 8 million documents and 40GB of text. We hold out a small portion of WebText as a validation set and select the checkpoint with the lowest validation loss. The exact training time and cluster configuration are not reported in this technical report.